THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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CLASS 9 — MATHS · NCERT-TYPE CHAPTER TEST SET 2

Chapter 14: **Surface Area and Volume** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Take $\pi = \frac{22}{7}$. The curved surface area of a cone of radius 7 cm and slant height 25 cm is: 1

(a) 662 cm^2 **(b)** 550 cm^2 **(c)** 606 cm^2 **(d)** 494 cm^2

2. Take $\pi = \frac{22}{7}$. The volume of a cone of radius 7 cm and height 24 cm is: 1

(a) 1108 cm^3 **(b)** 1232 cm^3 **(c)** 1356 cm^3 **(d)** 1480 cm^3

3. Take $\pi = \frac{22}{7}$. The surface area of a sphere of radius 10.5 cm is: 1

(a) 1386 cm^2 **(b)** 1247 cm^2 **(c)** 1525 cm^2 **(d)** 1665 cm^2

4. Take $\pi = \frac{22}{7}$. The volume of a sphere of radius 10.5 cm is: 1

(a) 5337 cm^3 **(b)** 4365 cm^3 **(c)** 4851 cm^3 **(d)** 5823 cm^3

5. Take $\pi = \frac{22}{7}$. The curved surface area of a hemisphere of radius 14 cm is: 1

(a) 1232 cm^2 **(b)** 1480 cm^2 **(c)** 1356 cm^2 **(d)** 1108 cm^2

6. The total surface area of a closed cylinder of radius r and height h is: 1

(a) $\pi r(r+h)$ **(b)** $2\pi rh$ **(c)** $2\pi r(r+h)$ **(d)** $\pi r^2 h$

7. Take $\pi = \frac{22}{7}$. The curved surface area of a cylinder of radius 3.5 cm and height 9 cm is: 1

(a) 198 cm^2 **(b)** 218 cm^2 **(c)** 239 cm^2 **(d)** 178 cm^2

8. Take $\pi = \frac{22}{7}$. The total surface area of a cylinder of radius 3.5 cm and height 29 cm is: 1

(a) 643 cm^2 **(b)** 860 cm^2 **(c)** 715 cm^2 **(d)** 787 cm^2

9. Take $\pi = \frac{22}{7}$. The curved surface area of a cone of radius 17.5 cm and slant height 45.5 cm is: 1

(a) 2251.5 cm^2 **(b)** 2753.5 cm^2 **(c)** 2502.5 cm^2 **(d)** 3004.5 cm^2

10. Take $\pi = \frac{22}{7}$. The volume of a cone of radius 42 cm and height 56 cm is: 1

(a) 103488 cm^3 **(b)** 124187 cm^3 **(c)** 93139 cm^3 **(d)** 113837 cm^3

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Take $\pi = \frac{22}{7}$. Find the curved surface area, total surface area of a solid hemisphere of radius 14 cm. 2

12. A pyramid has a square base of side 3 cm and height 36 cm. Find its volume. 2

13. Take $\pi = \frac{22}{7}$. A cone has radius 28 cm and height 52.5 cm. Find its slant height, curved surface area and volume. 2

14. Take $\pi = \frac{22}{7}$. Find the curved surface area, total surface area of a solid hemisphere of radius 7 cm. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Take $\pi = \frac{22}{7}$. A cone has radius 42 cm and height 56 cm. Find its slant height, curved surface area and volume. 3

16. Take $\pi = \frac{22}{7}$. Find the curved surface area, total surface area and volume of a solid hemisphere of radius 31.5 cm. 3

17. Take $\pi = \frac{22}{7}$. A cylindrical pillar has radius 21 cm and height 10 cm. Find the cost of painting its curved surface at ₹30 per 1000 cm^2. 3

OR

Take $\pi = \frac{22}{7}$. A cone has radius 21 cm and height 28 cm. Find its slant height, curved surface area and volume. 3

18. Take $\pi = \frac{22}{7}$. Find the curved surface area, total surface area and volume of a solid hemisphere of radius 42 cm. 3

OR

Take $\pi = \frac{22}{7}$. A cylindrical pillar has radius 14 cm and height 30 cm. Find the cost of painting its curved surface at ₹35 per 1000 cm^2. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. A rectangular sheet 88 cm long and 5 cm wide is rolled so that its length forms the circumference of the base of a cylinder of height 5 cm. Find the radius, the curved surface area and the volume (Take $\pi = \frac{22}{7}$). 5

OR

A solid metal sphere of radius 3 cm is melted and recast into small solid spheres of radius 1 cm. Find the number of small spheres and compare the total surface area of the small spheres with the surface area of the big sphere. 5

20. A rectangular sheet 88 cm long and 8 cm wide is rolled so that its length forms the circumference of the base of a cylinder of height 8 cm. Find the radius, the curved surface area and the volume (Take $\pi = \frac{22}{7}$). 5

OR

A solid metal sphere of radius 12 cm is melted and recast into small solid spheres of radius 3 cm. Find the number of small spheres and compare the total surface area of the small spheres with the surface area of the big sphere. 5

END OF QUESTION PAPER — Total: 40 Marks